

Large Language Models for solving mathematical problems

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Motivation

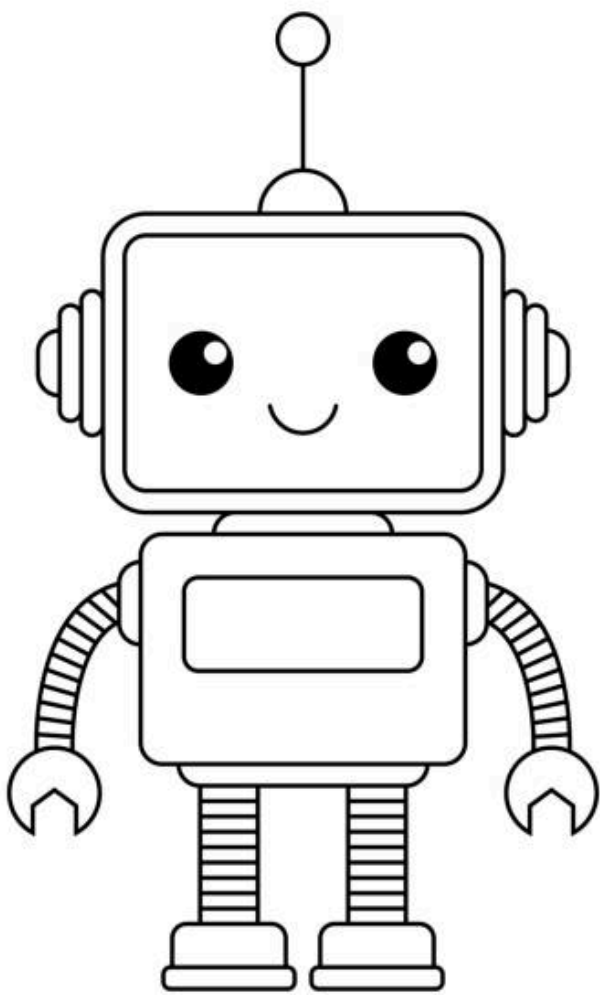
Large language models (LLMs) achieve impressive results across many language tasks, yet mathematical reasoning remains a persistent challenge. Models are trained on general web data not focused on structured, multi-step thinking. This work fine-tunes three open-source 7B models using Low-Rank Adaptation (LoRA) on the GSM8K dataset.

Dataset: **GSM8K**

- 8,500 grade school math problems created by human writers
- **Split:** 7,473 train / 1,319 test examples
- Chain-of-thought format with ##### answer delimiter
- Fewer than 2% of problems contain errors

Models

- **Llama-2-7b-hf** — base model, no instruction tuning
- **Llama-2-7b-chat-hf** — chat model (SFT + RLHF)
- **Mistral-7B-Instruct-v0.1** — instruct model (SFT only)



Experiments

A systematic hyperparameter search was conducted across 4 experiments:

- Experiment 1: Initial search over learning rate and LoRA rank
- Experiment 2: Effective batch size ablation
- Experiment 3: Fine-grained learning rate search
- Experiment 4: LoRA rank and alpha ablation

Evaluation uses **exact-match accuracy**.

Results

Model	Baseline	Fined-tuned	Gain
Llama-2-7b-hf	10.77%	33.36%	+22.6%
Llama-2-7b-chat-hf	25.47%	39.88%	+14.4%
Mistral-7B-Instruct-v0.1	42.23%	57.01%	+14.8%

Conclusions and Future work

- LoRA fine-tuning significantly improves mathematical reasoning in all three models
- Pre-training type determines the hyperparameter configuration
- Future: test other datasets and compare with QLoRA / full fine-tuning